Software Requirement Specification (SRS)

Project: AI-Powered Car Diagnostic Mobile Application

Version: 1.0
Status: Final Draft for Stakeholder Review
Role: Business Development / Product Management

1. Introduction

1.1 Purpose

The purpose of this document is to provide a detailed overview of the software requirements for the AI-Based Car Diagnostic Mobile Application. This tool is designed to assist vehicle owners in identifying mechanical issues through visual and auditory data, facilitating safer driving and informed maintenance.

1.2 Project Scope

The application will use smartphone sensors (Camera and Microphone) to analyze vehicle health. It acts as an intermediary tool that provides preliminary diagnostics and basic repair tutorials while bridging the communication gap between car owners and professional mechanics.

1.3 Definitions and Acronyms

- **OCR:** Optical Character Recognition (used for dashboard lights).
- **ANN:** Artificial Neural Networks (used for sound classification).
- **MVP:** Minimum Viable Product.
- **Offline Mode:** Data processing occurring locally on the device without cloud dependency.

2. Overall Description

2.1 User Classes and Characteristics

- **General Car Owners:** Users with little to no mechanical knowledge who need simplified explanations.
- **Mechanics/Technicians:** Professional stakeholders who provide the validated dataset for AI training and receive "pre-diagnosed" reports from users.

2.2 Design and Implementation Constraints

- **Hardware:** Must run on mid-range smartphones with standard camera and microphone quality.
- **Environment:** The audio diagnostic tool must account for ambient noise (traffic, wind, rain).
- **Language:** Full bilingual support (English and French).

3. System Features (Functional Requirements)

3.1 Visual Diagnostic Module (Dashboard Recognition)

- **Description:** The system shall capture images of the car dashboard and identify active warning lights.
- **Functional Requirement (FR-1):** The app must categorize identified lights into urgency levels: **Red** (Stop immediately), **Yellow** (Service required soon), and **Green/Blue** (Information).
- **FR-2:** The app must provide a plain-language explanation of each icon.

3.2 Acoustic Diagnostic Module (Engine Sound Mapping)

- **Description:** The system shall record engine sounds for a duration of 10–15 seconds to identify anomalies.
- **FR-3:** The system must match recorded frequencies against a validated database of mechanical faults (e.g., worn belts, misfiring pistons, exhaust leaks).
- **FR-4:** The system shall display a "Confidence Score" for each sound-based diagnosis to manage user expectations.

3.3 Maintenance & Tutorial Module

- **Description:** A library of instructional content for non-critical repairs.

- **FR-5:** The system shall provide step-by-step video and text tutorials for basic tasks (e.g., checking oil, changing a tire, jump-starting a battery).
- **FR-6:** The system shall flag "Critical Failures" where DIY repair is prohibited for safety reasons, directing the user to the nearest mechanic.

3.4 History and Report Generation

- **FR-7:** The system shall store a log of all scans, including timestamps, detected faults, and recommended actions.

4. Non-Functional Requirements

4.1 Reliability and Accuracy

- **NFR-1:** Visual recognition must achieve a minimum accuracy of 95% in varying lighting conditions.
- **NFR-2:** Sound diagnostics must prioritize "Safety-First" logic, defaulting to a professional inspection recommendation if the AI confidence score is below 70%.

4.2 Performance & Availability

- **NFR-3 (Offline Capability):** The core diagnostic engine must be stored locally on the device to allow functionality in areas with zero or low internet connectivity.
- **NFR-4:** Image/Sound processing should not exceed 10 seconds per scan on standard hardware.

4.3 Usability

- **NFR-5:** The user interface must utilize icons and minimal text to ensure accessibility for users with varying literacy levels.

5. Dependency Relationships & Priorities

ID	Requirement	Priority	Dependencies
01	Dashboard Light Recognition	**P1 (Critical)**	Camera API, OCR Engine
02	Offline Local Database	**P1 (Critical)**	Storage Optimization
03	Engine Sound	**P2 (High)**	Microphone API, Audio

ID	Requirement	Priority	Dependencies
	Mapping		Training Set
04	Bilingual Support	**P2 (High)**	Translation Database
05	Video Tutorials	**P3 (Medium)**	External Content/Local Compression

6. Technical Feasibility Analysis

- **Feasible:** Dashboard recognition and offline database management are highly feasible using existing frameworks (TensorFlow Lite, OpenCV).
- **Challenging:** Acoustic diagnostics in real-world environments require significant noise-cancellation filtering and a large, diverse dataset of recorded engine sounds to avoid false positives.

7. Conclusion

This SRS outlines a robust framework for a tool that addresses specific local challenges: high mechanical ignorance and inconsistent connectivity. By focusing on the "Accuracy-Trust Paradox," the development will prioritize the P1 requirements to ensure the MVP is both reliable and functional in the field.

Approval Sign-off: *Business Development Lead Technical Project Manager*